GENERATIVE ARTIFICIAL INTELLIGENCE

Store Operations Analytics Dashboard for DVD Rental Operations

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ABSTRACT

This report details the development of the Store Operations Analytics Dashboard, an integrated business intelligence system designed to optimize decision-making for a dual-store DVD rental business. Built using Python, Streamlit, and PostgreSQL, the dashboard transforms over 16,000 transactional records into a multi-layered analytical experience.

The system is structured into four primary modules: Executive Overview for high-level KPIs; Revenue Pulse for temporal trend analysis and store to store comparisons; Film & Inventory Intelligence for category performance and pricing strategy; and the Business Decision Simulator (Leaks Analysis), which utilizes predictive modeling to identify revenue leakage from late returns and simulate recovery scenarios. By leveraging scikit-learn for revenue forecasting and Plotly for interactive visualization, this dashboard provides store managers with both historical clarity and predictive foresight to enhance operational efficiency.

INTRODUCTION

1.1 Background

In the era of big data, the ability to extract meaningful narratives from relational databases is a competitive necessity. The DVDRental database, while rich in transactional history, remains a "black box" for operational managers without technical SQL skills. This project aims to bridge that gap by creating a user-centric dashboard that monitors the vital signs of store operations in real-time.

1.2 Problem Statement

Managing a retail rental business involves complex variables that are difficult to track manually:

- Operational Blind Spots: Managers often lack clarity on which store (Store 1 vs. Store 2) is performing more efficiently across different timeframes.
- Inventory Mismanagement: Without genre-based revenue analysis, it is difficult to identify "Best Sellers" versus "Dead Stock."
- Revenue Leakage: Late returns and unoptimized pricing represent "leaks" in the business model that are often hidden in raw data.
- Lack of Predictive Tools: Traditional reports only show the past, leaving managers without a way to simulate the impact of future policy changes.

1.3 Project Objectives

The primary goals of this development are:

1. Metric Centralization: To provide a single source of truth for KPIs including Total Revenue ($61,312), Total Payments (14,596), and Active Customers (1,198).
2. Temporal Insight: To identify the "Revenue Pulse" and peak rental periods (e.g., Mondays and Fridays) to optimize staffing.
3. Inventory Optimization: To analyze film performance by category and pricing to maximize turnover.

4. Leakage Recovery: To deploy a Business Decision Simulator that quantifies the financial impact of operational improvements, specifically in reducing late returns.

1.4 Scope of Work

The scope encompasses the full data lifecycle, including:

- Backend: Designing complex SQL views in PostgreSQL for aggregated analytics.
- Data Science: Implementing a Gradient Boosting Regressor for revenue forecasting.
- Frontend: Developing an interactive, dark-themed Streamlit application with real-time filtering for stores and time periods.
- Prescriptive Analytics: Creating simulation tools to model "What-If" business scenarios.

SYSTEM ARCHITECTURE

The Store Operations Analytics Dashboard is engineered using a robust, modular multi-tier architecture. This design ensures a clean separation between data management, statistical processing, and the user interface, allowing for high performance even when handling thousands of transactional records.

2.1 Data Layer (Database & Connectivity)

- Source Database: The system utilizes the PostgreSQL (dvdrental) database as the primary data source, containing complex relational tables for rentals, payments, and inventory.
- Database Connector (db.py): Connectivity is managed via Psycopg2 and SQLAlchemy. This layer includes a centralized run_query function with @st.cache_data implementation to minimize database load and ensure near-instant response times during repeated filtering.
- Feature Engineering: The system uses SQL-level aggregations to calculate critical business metrics such as rental duration days and month-over-month revenue growth before the data enters the Python environment.

2.2 Logic & Transformation Layer (The Engine)

- Main Orchestrator (main_project.py): Acts as the entry point, managing global configurations, high-end CSS injection for the "Calm Dark" professional theme, and the multi-page navigation system.
- Data Manipulation: Pandas is used extensively across all modules (Overview, Revenue, Film, and Leaks) to perform real-time data slicing, filtering by Store ID, and recalculating KPIs based on user-selected date ranges.

2.3 Intelligence & Predictive Layer

- Model Training (train_model.py): A dedicated pipeline that trains a Random Forest Regressor to understand the relationship between rental rates, film categories, and return velocities.
- Machine Learning Models (model.pkl & label_encoder.pkl): The dashboard integrates pre-trained scikit-learn models to provide "Intelligence" features.
    - Revenue Forecasting: Uses Linear Regression (in 2_revenue_stores.py) for trend-based financial projections.

    - Duration Prediction: Uses the Random Forest model to predict film return velocity based on category and pricing.
- Prescriptive Logic (4_leaks_action.py): A custom simulation engine that calculates "Potential Late Fees" and "Revenue Recovery" using real-time user inputs from the simulation sliders.

2.4 Presentation Layer (User Interface)

- Streamlit Frontend: A reactive web framework that serves as the interface for both technical and non-technical stakeholders.
- Interactive Visualizations: Powered by Plotly, providing dynamic Line Pulses, Grouped Bar Charts, and Sunburst/Donut charts for category intelligence.
- UI Components: Custom helper functions (kpi, summary, sec) ensure a consistent and professional visual hierarchy throughout the application.

ANALYTICAL MODULES

The dashboard is structured into several integrated analytical modules, each designed to answer specific business questions and support decision-making from different operational perspectives. By combining descriptive SQL aggregations with predictive machine learning, the system transforms raw data into actionable intelligence.

3.1 Executive Overview: The Temporal Revenue Pulse

The Overview module serves as the entry point of the dashboard, providing a high-level summary of overall business performance. It is designed to give users a quick yet comprehensive understanding of the current operational condition before moving into deeper analysis.

- Key Metrics Aggregation: Utilizing the kpi() helper function, this module aggregates vital signs such as Total Revenue ($61,312), Total Transactions (14,596), and Unique Active Customers (1,198). These are displayed in a compact format to help users quickly assess business scale and efficiency.
- Monthly Revenue Snapshot: Visualizes revenue trends over time using grouped bar charts. This enables users to identify patterns such as growth, decline, or fluctuations, as well as evaluate whether performance is stable or changing.
- Store Snapshot: Provides a direct comparison of revenue between branches. This allows for quick benchmarking, helping users detect even small performance differences between Store 1 and Store 2 that may require further investigation.

3.2 Revenue and Store Performance

This module focuses on analyzing revenue trends over time while also comparing performance between locations.

- Time-Series Evolution: Through the "Revenue Pulse" line chart, users can observe how revenue evolves across different granularities (Daily, Weekly, Monthly). This helps stakeholders identify growth patterns and detect temporal performance gaps.
- Store Battle Engine: A specialized comparative logic identifies which store performs better and quantifies the revenue gap. This provides insight into market share distribution between the two physical locations.

- Next-Month Forecasting: Enhanced with a predictive component using Linear Regression, the system estimates next-month revenue based on historical trends, aiding in high-level strategic planning and goal setting.

3.3 Film and Pricing Intelligence

This module explores product-level performance by examining how different film categories and pricing strategies impact revenue.

- Category-Based Revenue: Identifies which genres (e.g., Sports, Sci-Fi) contribute the most to business success, revealing deep customer preferences.
- Return Velocity Prediction: Utilizing a Random Forest Regressor (trained with 150 estimators in train_model.py), the system predicts the "Estimated Return Duration" for films. This allows managers to understand inventory turnover cycles.
- Pricing Strategy Optimization: By analyzing rental rates against category demand, the module helps identify whether certain films are underpriced or overpriced, enabling managers to optimize inventory allocation and price points.

3.4 Revenue Leakage and Recovery

Designed to uncover operational inefficiencies, this module highlights "hidden" losses that are often overlooked in traditional reports.

- Quantifying Inefficiencies: Instead of only focusing on recorded revenue, the module calculates Potential Late Fees, revenue lost or delayed due to late returns. By comparing actual revenue with potential fees, the system quantifies the "leaks" in the current business model.
- Late Return Rate Analysis: The system identifies categories or stores with high late-return frequencies, highlighting operational gaps in fee enforcement.
- Business Decision Simulator: An interactive tool where managers can simulate the financial impact of operational improvements. For example, reducing late returns via a slider shows an immediate estimated revenue uplift, transforming the dashboard from a passive reporting tool into an active decision-support system.

3.5 AI Revenue Forecasting with Transformer Model

This module is designed to predict future business revenue using an AI-based Transformer forecasting model. Unlike traditional reporting systems that only display historical data, this module provides predictive analytics to help management anticipate future business performance and support strategic decision-making.

- Transformer-Based Forecasting: The system uses a Vanilla Transformer Encoder architecture built with PyTorch. Historical monthly revenue data is processed using sliding-window sequences, where the model learns patterns from previous months and predicts revenue for upcoming periods.
- Positional Encoding Implementation: To improve sequence understanding, the model applies sinusoidal positional encoding. This enables the Transformer to recognize temporal relationships between historical revenue points, improving forecasting accuracy for time-series data.

- Forecast with Confidence Interval: The module does not only generate revenue predictions, but also provides confidence intervals using Monte Carlo Dropout. By running multiple prediction passes, the system calculates upper and lower prediction bounds, helping users understand forecasting uncertainty and risk levels.
- Automatic Fallback Forecasting: If PyTorch is unavailable, the module automatically switches to fallback forecasting methods such as Exponential Smoothing or Moving Average Trend analysis. This ensures the forecasting feature remains functional even in limited deployment environments.
- Caching and Performance Optimization: To reduce repeated model training and improve system performance, the module implements an in-memory caching mechanism. Previously generated forecasts are temporarily stored and reused within a configurable time period.
- Business Decision Support: The forecasting results can help managers estimate future revenue trends, identify potential business decline risks, and prepare operational strategies proactively. This transforms the dashboard from a passive reporting platform into an intelligent AI-driven business decision support system.

3.5 AI-Based Dashboard Customization Feature

One of the latest improvements added to the *Store Operations Analytics Dashboard* is the implementation of an AI-based dashboard customization feature. This feature was developed to allow users to modify the dashboard appearance directly based on their preferences without manually editing the source code.

Unlike traditional dashboards that only store temporary changes during a session, this system is designed to save all modifications permanently. This means that when users change dashboard colors, chart styles, or layout positions, the changes will remain even after refreshing the page or reopening the application.

To support this functionality, the system uses an integrated configuration storage mechanism. Every customization made by the user is automatically updated and saved into the system configuration, making the dashboard more flexible and easier to personalize.

Developed Features

Dashboard Appearance Customization

Users are able to modify several visual elements of the dashboard, including:

- background colors,
- text colors,
- dashboard titles,
- and other interface components.

These changes are not temporary, as the system automatically stores them permanently so the dashboard will continue using the updated appearance in future sessions.

Dynamic Chart and Color Modification

The dashboard also supports dynamic data visualization adjustments. Users can change chart types such as:

- line charts,
- bar charts,
- pie charts,
- donut charts,
- and area charts.

In addition, chart colors can also be customized to create a more attractive and user-friendly analytical interface.

More Interactive Dashboard Navigation

The system includes an interactive navigation feature that allows users to directly access specific pages or charts through *Quick Actions* buttons. This feature improves usability and makes data exploration faster and more efficient.

Natural Language Commands

One of the main highlights of this development is the integration of Generative AI to process commands written in natural language. Users can simply type instructions such as:

- "change the dashboard theme to neon blue"
- "convert the revenue chart into a pie chart"
- "move the forecasting graph to the top section"

The AI system will then process the instruction and automatically apply the requested modifications to the dashboard.

Flexible Layout Arrangement

The dashboard also supports layout customization, allowing users to rearrange charts and widgets according to their analytical preferences. This feature makes the dashboard feel more dynamic and adaptable for different business needs.

PREDICTIVE MODELING & MACHINE LEARNING

The dashboard moves beyond static analysis by allowing users to actively explore future outcomes through two distinct predictive approaches: Time-Series Forecasting and Behavioral Simulation.

4.1 Predictive Modeling Approach (Module: Revenue Pulse)

The system incorporates a simple yet effective time-series forecasting method to assist in financial planning and goal setting.

- Algorithm: Moving Average (MA3) & Linear Trendline.
- How it works: * The model calculates the average revenue of the last 3 periods to smooth out fluctuations.
    - This data is then used to project the next month’s revenue, visualized as a dashed line extending from the latest historical data point in the Revenue Pulse chart.
- Why this approach: * Data Suitability: Highly effective for datasets with limited time-frames where complex models might overfit.
    - Interpretability: Provides a clear "baseline" for business users to gauge if current performance is above or below historical averages.

4.2 Pricing Simulation Model (Module: Film & Pricing)

To understand the relationship between pricing and inventory turnover, the system utilizes a Random Forest Regression model (implemented in train_model.py and loaded via model.pkl).

- Algorithm: Random Forest Regressor.
- How it works:
    - Training: The model is trained on historical rental transactions, specifically looking at the rental_rate, category (encoded), and store_id.
    - Feature Engineering: Categorical data is transformed using LabelEncoder to ensure the model can process genre-specific impacts.
    - Interaction: Users can interact with the simulator by selecting a specific Store, choosing a Film Category, and adjusting the Rental Price slider.
    - Prediction: The model generates a real-time prediction of the Rental Duration (duration_days), reflecting how long a customer is expected to hold a film under those specific conditions.
- Why Random Forest:
    - Non-Linear Capture: It effectively captures complex customer behaviors that simple statistics might miss (e.g., how a price hike affects a "Documentary" differently than a "New Release").
    - Flexibility: It is robust against outliers and provides more realistic "What-If" scenarios for interactive simulations.

4.3 Revenue Leakage & Recovery Simulation (Module: Leaks Action)

This prescriptive logic (found in 4_leaks_action.py) works as a strategic layer over the predictive models to identify and recover lost income.

- Logic: Prescriptive Impact Simulation.
- Purpose: To quantify "Potential Gains" by identifying revenue leakage from late returns and uncollected fees.
- How it works:
    - The system identifies hidden losses by comparing actual revenue with potential late fees.

- Through an Interactive Simulation Slider, managers can model the financial impact of reducing these "leaks."
- Strategic Outcome: This enables users to identify optimal price points that balance revenue and inventory turnover, while providing actionable recommendations to improve fee enforcement.

4.4 Persistent Configuration Architecture

To ensure that all dashboard modifications remain permanent, the system implements a *Persistent Configuration Architecture*. With this approach, all dashboard settings are stored in configuration files so they are not lost when the application is refreshed or restarted.

Several core components are used in this architecture:

Configuration Manager

Responsible for reading and saving all dashboard configurations modified by users.

Dashboard Renderer

Handles the automatic rendering of the dashboard based on the latest saved configurations so every change can immediately appear on the interface.

Theme Management System

Controls dashboard appearance elements such as colors, typography, and chart styling to maintain visual consistency.

Layout Management

Manages chart and widget positioning, allowing users to customize dashboard arrangements more flexibly.

AI Command Processor

This component processes user instructions written in natural language and translates them into direct dashboard modifications.

CONCLUSION

The Store Operations Analytics Dashboard successfully transforms the complex “DVDRental” dataset into a powerful and user-centric decision support system.By integrating descriptive analytics with machine learning models, specifically Random Forest for inventory velocity prediction and Linear Regression for revenue forecasting, the project bridges the gap between raw operational data and strategic business decision-making.

The implementation of the Business Decision Simulator further enhances the platform by transforming it from a standard reporting dashboard into a more prescriptive analytical system. Through this feature, managers are able to identify revenue leakage, evaluate operational inefficiencies, and simulate recovery strategies in real time.

In addition, the integration of Generative Artificial Intelligence introduces a more adaptive and interactive dashboard experience. Users are able to customize dashboard themes, modify chart structures, rearrange layouts, and navigate analytical pages using natural language commands. Unlike conventional dashboards where changes are temporary, this system stores user customizations permanently, creating a more flexible and personalized analytical environment.

Overall, the dashboard not only improves operational visibility and analytical efficiency, but also demonstrates how AI-driven technologies can enhance modern business intelligence systems. With its combination of predictive analytics, interactive visualization, and self-customizing capabilities, the platform provides a scalable and innovative solution for supporting data-driven decision making in retail business operations.